Student Spending Behavior & Lifestyle Survey

Kollati Devi Chandrika
Indian Institute of Technology Bhilai
Bhilai, Chhattisgarh, India
kollatidc@iitbhilai.ac.in

Abstract

This study analyzes student spending behavior and lifestyle patterns using survey data. The objective is to understand how students allocate expenses across categories such as food, transportation, entertainment, and academic needs. Statistical and exploratory data analysis techniques are applied to identify trends, relationships, and key factors influencing spending habits. The results provide insights into financial behavior among students and highlight how lifestyle choices impact expenditure patterns. These findings can support better financial planning and decision-making for students.

1 Introduction

1.1 Background

Student spending behavior plays a crucial role in understanding financial habits among young adults. With limited financial resources, students must balance academic expenses, daily necessities, and lifestyle choices such as entertainment and eating out. These decisions significantly influence their overall financial stability.

In recent years, changing lifestyles and increased accessibility to food delivery, online shopping, and social activities have impacted how students manage their expenses. Therefore, analyzing spending behavior has become important for understanding financial discipline and decision-making patterns among students.

1.2 Problem Statement

Despite having limited income or allowance, students often exhibit inconsistent spending and saving patterns. Lifestyle factors such as eating-out frequency and personal habits may significantly influence their expenditure.

This study aims to analyze and quantify the relationship between student spending, savings, and lifestyle behaviors. It also seeks to identify key factors that drive higher spending and understand how financial habits vary among individuals.

2 Objectives

- To collect and preprocess survey-based data on student spending behavior
- To analyze relationships between spending, savings, and lifestyle habits
- To visualize patterns using Exploratory Data Analysis (EDA)
- To identify key factors influencing student expenditure
- To develop a predictive model for estimating student spending

3 Dataset Description

The dataset used in this study consists of survey responses collected from students to understand their spending behavior and lifestyle patterns. The data captures various aspects of financial activity, including monthly expenditure, savings, and lifestyle choices.

The dataset includes the following features:

- Demographic Features: Gender of the respondents
- Financial Metrics: Monthly spending and savings
- Lifestyle Metrics: Frequency of eating out

These features provide insights into how students manage their finances and how lifestyle factors influence their spending behavior.

4 Methodology

The study follows an Exploratory Data Analysis (EDA) approach to understand patterns in student spending behavior. The analysis pipeline includes the following steps:

- Data cleaning and preprocessing to ensure consistency and completeness
- Descriptive statistical analysis to understand central tendencies and distributions
- Feature-wise exploration of spending categories and lifestyle attributes
- Data visualization using histograms, bar charts, scatter plots, and correlation heatmaps
- Identification of relationships between spending behavior and lifestyle factors

The analysis was implemented using Python, leveraging libraries such as Pandas for data manipulation, NumPy for numerical computation, and Matplotlib and Seaborn for visualization.

5 Data Cleaning and Preparation

Real-world survey data often contains inconsistencies, missing values, and irregular formats. Therefore, several preprocessing steps were applied to ensure data quality and reliability.

5.1 Handling Missing Values

Empty rows and missing values were removed to maintain dataset consistency and accuracy.

5.2 Data Transformation

Text-based financial entries such as “2000–3000” were converted into numerical values by extracting numbers and computing their average. Currency symbols, commas, and spaces were removed to standardize the data.

5.3 Feature Engineering

New numerical features were created from textual data:

- Spending_num: Numeric representation of monthly spending
- Savings_num: Numeric representation of savings

- EatOut_num: Encoded frequency of eating out

5.4 Categorical Encoding

Categorical variables such as gender were encoded into numerical format to facilitate analysis.

5.5 Duplicate Removal

Duplicate records were removed to ensure data integrity using standard data cleaning techniques.

6 Exploratory Data Analysis and Visualization

To understand student spending behavior, a set of visualizations was created. Each plot highlights key relationships between financial and lifestyle variables.

6.1 Spending Distribution by Gender

The figure below shows the distribution of spending across gender.

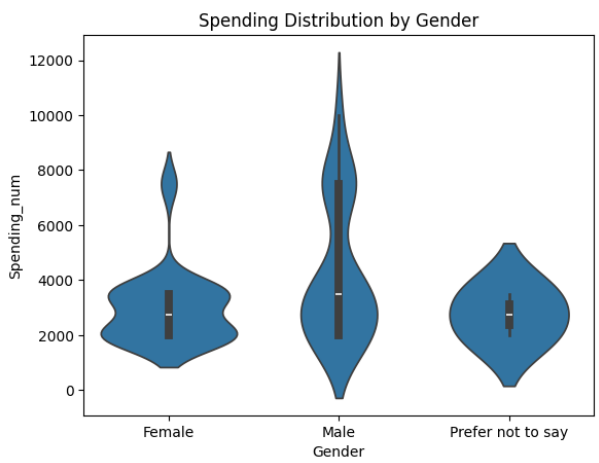

Figure 1: Spending Distribution by Gender

Insight: Spending patterns vary across gender, with noticeable differences in variability. While most students fall within a moderate spending range, certain groups exhibit wider dispersion, indicating diverse financial behavior and spending habits.

6.2 Spending vs Savings Relationship

The scatter plot illustrates the relationship between spending and savings.

Insight: The plot shows no strong linear relationship between spending and savings, suggesting that these two factors are not directly dependent. Some students are able to maintain savings despite higher spending, indicating differences in financial planning and discipline.

6.3 Impact of Eating Out on Spending

The regression plot shows the effect of lifestyle habits on spending.

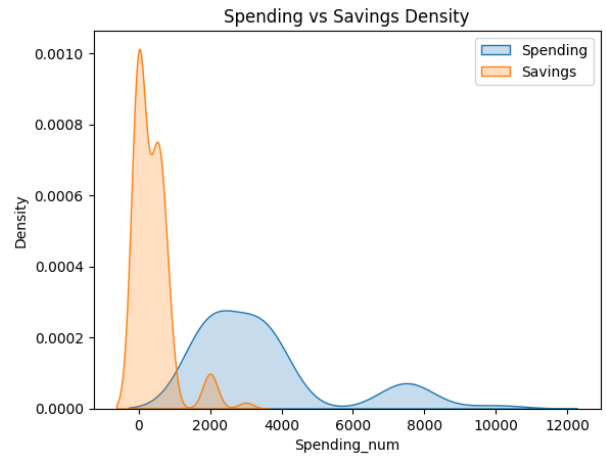

Figure 2: Relationship between Spending and Savings

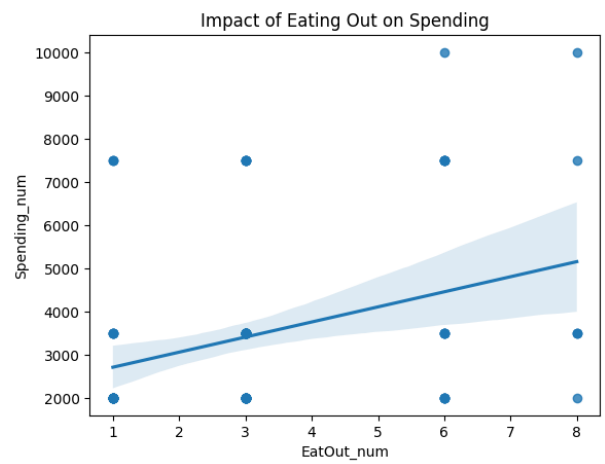

Figure 3: Impact of Eating Out on Spending

Insight: A clear positive trend is observed between eating-out frequency and spending. This indicates that lifestyle choices significantly influence financial behavior, with frequent dining out contributing to higher overall expenditure.

6.4 Correlation Heatmap

The heatmap provides a summary of relationships between key variables.

Insight: The heatmap confirms a positive correlation between eating-out frequency and spending, while savings shows only a weak relationship with other variables. This suggests that lifestyle factors play a more dominant role in influencing spending behavior than saving habits.

6.5 Distribution of Spending

The distribution plot shows overall spending behavior among students.

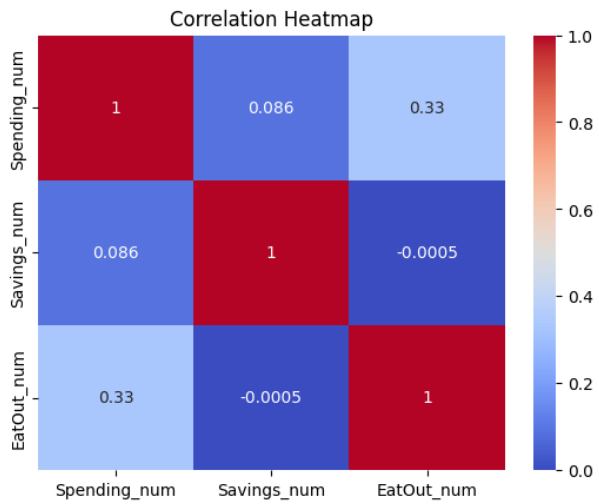

Figure 4: Correlation Matrix of Key Variables

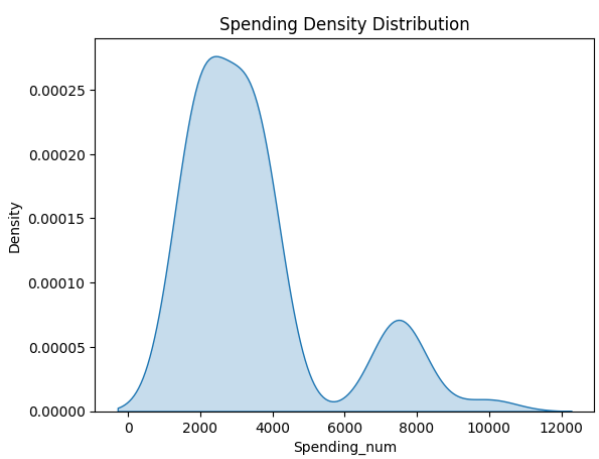

Figure 5: Distribution of Student Spending

Insight: The distribution indicates that most students fall within a mid-range spending category, reflecting consistent spending behavior. The slight right skew suggests the presence of a smaller group of high spenders with significantly higher expenditure.

6.6 Summary of Insights

The visualizations collectively show that student spending behavior is primarily influenced by lifestyle choices rather than savings patterns. Eating-out frequency emerges as a key factor driving higher expenditure, while financial discipline varies across individuals. Overall, most students demonstrate moderate spending with some variation across the dataset.

7 Model Building

7.1 Objective

The objective of this model is to predict student spending behavior based on financial and lifestyle factors such as savings, eating-out frequency, and gender.

7.2 Model Selection

A Decision Tree Regressor was selected for this study due to its ability to capture non-linear relationships and interactions between variables. It is also easy to interpret and works well with structured survey data.

7.3 Feature Selection

The following features were used as input variables for the model:

- Savings_num
- EatOut_num
- Gender_num

The target variable used for prediction is:

- Spending_num

7.4 Model Implementation

The dataset was divided into training and testing sets using an 80-20 split. The Decision Tree model was trained on the training data with parameters such as maximum depth and minimum samples per split to reduce overfitting. Predictions were then generated on the test dataset.

7.5 Model Evaluation

The model was evaluated using Mean Absolute Error (MAE) and the coefficient of determination (R^2 score).

- MAE: [1400.33]
- R^2 Score: [0.256]

The MAE represents the average prediction error, while the R^2 score indicates how well the model explains the variability in student spending.

7.6 Actual vs Predicted Plot

Insight: The plot shows how close the predicted values are to the actual spending values. The spread indicates moderate model accuracy, with some deviations from the ideal prediction line.

7.7 Residual Plot

Insight: Residuals are distributed around zero, but with noticeable spread. This suggests that the model has prediction errors and does not fully capture all influencing factors.

7.8 Interpretation

The model results indicate that eating-out frequency has a strong influence on student spending, confirming the trends observed during exploratory analysis. Savings show a weaker relationship, suggesting that spending decisions are not strictly dependent on saving behavior. Gender contributes slightly to variation but is not a dominant factor.

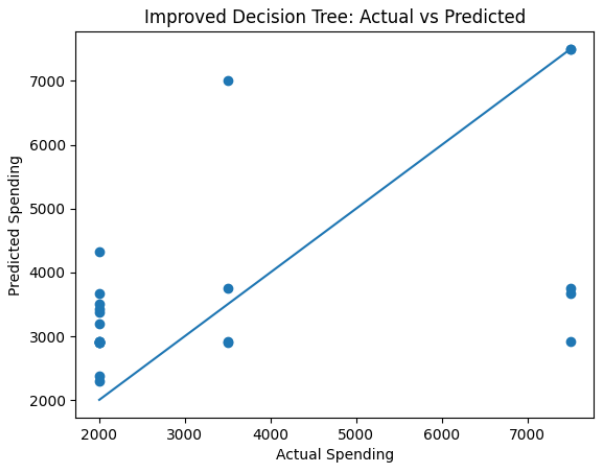

Figure 6: Actual vs Predicted Spending

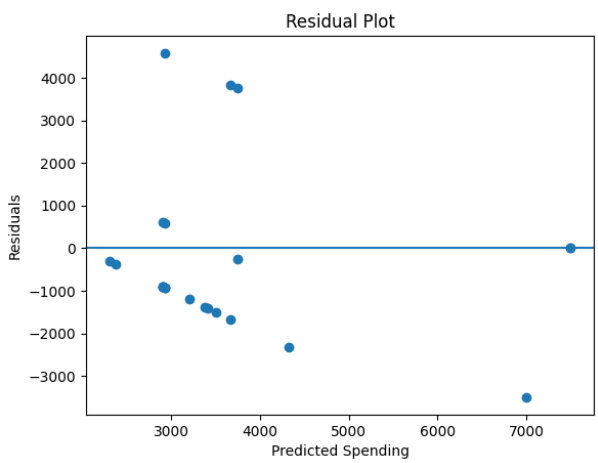

Figure 7: Residual Plot

Overall, the model captures general spending patterns but may not fully explain individual differences due to limited features.

7.9 Limitations

The performance of the model is limited by the small number of input features and the variability in human financial behavior. Important factors such as income level, personal preferences, and external influences are not included in the dataset.

8 Results and Discussion

The exploratory analysis reveals clear patterns in student spending behavior based on financial and lifestyle variables. A significant portion of monthly expenditure falls within a moderate range, while variations are influenced by individual habits.

Key insights from the analysis include:

- Students who frequently eat outside tend to have higher overall spending, indicating a strong influence of lifestyle habits on expenditure.
- Spending values are more widely distributed compared to savings, suggesting that students prioritize spending over saving.
- There is no strong direct relationship between spending and savings, highlighting differences in personal financial management.
- Gender-based analysis shows some variation in spending distribution, although it is not a dominant factor.

The visualizations and model results collectively indicate that lifestyle choices, particularly eating-out frequency, play a more significant role in determining spending behavior than savings.

9 Conclusion

This study provides a comprehensive analysis of student spending behavior and lifestyle patterns using survey-based data. The findings indicate that spending habits are strongly influenced by lifestyle choices, especially eating-out frequency.

The analysis highlights that savings have a weaker relationship with spending, and financial behavior varies significantly across individuals. The predictive model was able to capture general trends but showed moderate performance, suggesting that additional factors may influence spending behavior.

These insights emphasize the importance of financial awareness among students and demonstrate how data-driven analysis can help understand real-world behavioral patterns. Future work can improve this study by incorporating additional features such as income, personal preferences, and broader lifestyle factors to enhance prediction accuracy.